

Conceptual Questions

1. What's the motivation behind measuring OFI at multiple depth levels of the order book?

As noted in the paper (2.1.3):

"There exist strong correlations between multi-level OFIs, and the first principal component can explain over 89% of the total variance among multi-level OFIs."

This suggests that measuring OFI at multiple depth levels might allow us to capture more information about supply and demand dynamics beyond just the best bid and ask.

2. Why do the authors use Lasso regression rather than OLS for estimating cross-impact?

The authors choose Lasso regression to deal with high-dimensionality and improve interpretability. In Section 3, they say:

"Given the potentially high-dimensional structure of cross-impact, we apply Lasso regression to avoid overfitting and enhance interpretability."

It seems to me Lasso is used here to select a smaller set of predictors, which makes interpreting model easier and less likely to overfit in a high-dimensional setting where predictors are correlated.

3. Why is OFI considered a better predictor of short-term returns than trade volume?

OFI is considered a better predictor of short-term returns than trade volume because it includes events such as limit orders and cancellations, not just trades. This gives a more complete view of the market. Furthermore, while trade volume treats all trades symmetrically, OFI is defined differently for buying and selling. This allows it to capture the direction of order flow and tells us whether there are more buying or selling activities over time.